

# CFD 2025-26 – Lab 06

## Exercise 1

Let us consider the following Navier-Stokes problem in the unit square  $\Omega = (0, 1)^2$ :

$$\begin{cases} \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} - \frac{1}{\text{Re}} \Delta \mathbf{u} + \nabla p = \mathbf{f}, & \text{in } \Omega, \quad \forall t \in (0, T), \\ \text{div } \mathbf{u} = 0, & \text{in } \Omega, \quad \forall t \in (0, T), \\ \mathbf{u} = \mathbf{u}_{ex}, & \text{on } \partial\Omega, \quad \forall t \in (0, T), \\ \mathbf{u} = \mathbf{0}, & \text{in } \Omega, \quad \text{for } t = 0, \end{cases}$$

where the forcing term  $\mathbf{f}$  is given by

$$\mathbf{f}(\mathbf{x}, t) = \begin{bmatrix} -2 \cos x \sin y \left( \cos(2t) + \frac{1}{\text{Re}} \sin(2t) \right) \\ 2 \sin x \cos y \left( \cos(2t) + \frac{1}{\text{Re}} \sin(2t) \right) \end{bmatrix},$$

and the exact solution  $(\mathbf{u}_{ex}, p_{ex})$  is defined by

$$\mathbf{u}_{ex}(\mathbf{x}, t) = \begin{bmatrix} -\cos x \sin y \sin(2t) \\ \sin x \cos y \sin(2t) \end{bmatrix}, \quad p_{ex}(\mathbf{x}, t) = -\frac{\cos(2x) + \cos(2y)}{4} \sin^2(2t).$$

- 1 Introduce the forward Euler method to discretize the problem in time. Then, introduce suitable finite element spaces and derive the fully discrete problem associated to a timestep advancement.
- 2 Starting from the provided template, implement the method obtained at point 1. What can you observe?
- 3 Modify the scheme by treating different terms in an implicit way: what can you observe?
- 4 Consider different timesteps  $\Delta t = 0.04 \cdot [1, 1/2, 1/4, 1/8]$ . For each value of  $\Delta t$ , compute the error for the velocity and the pressure using the norm of the spaces  $L^\infty(0, 1, H^1(\Omega))$  and  $L^\infty(0, 1, L^2(\Omega))$ , respectively. What can you conclude on the convergence rate of the method?

Hint: to reduce computational time, use the diagonal preconditioner  $\text{diag}(P_*)$ .

- 5 Repeat the exercise using the following time-advancing schemes:
  - two-step BDF semi-implicit scheme;
  - two-step BDF explicit scheme.

In both cases, use a step of the backward Euler method to initialize the solver, and a second-order linear extrapolation

$$\mathbf{u}^{n+1} \simeq 2\mathbf{u}^n - \mathbf{u}^{n-1}$$

to approximate the advection field.